Solution

Q1 a) $f_X(x;\theta) = \frac{1}{\sqrt{2\pi x^3 \theta}} e^{-\frac{(x-1)^2}{2x\theta}}, \quad x>0, \ \theta>0.$

$$L(\theta) = \prod_{i=1}^{n} f_X(x_i;\theta) = \prod_{i=1}^{n} \left[\frac{1}{\sqrt{2\pi x_i^3 \theta}} e^{-\frac{(x_i-1)^2}{2x_i\theta}} \right] \quad (1 \text{ mark})$$

$$= \left(\frac{1}{(2\pi\theta)^{n/2} \prod_{i=1}^{n} x_i^{3/2}} \right) \left(e^{-\frac{1}{2\theta}\sum_{i=1}^{n} \frac{(x_i-1)^2}{x_i}} \right)$$

$$= \underbrace{\left[\frac{e^{-1/2\theta \sum_{i=1}^{n} \frac{(x_i-1)^2}{x_i}}}{(2\pi\theta)^{n/2}} \right]}_{g(T(\underline{x});\theta)} \underbrace{\left[\frac{1}{\prod_{i=1}^{n} x_i^{3/2}} \right]}_{h(\underline{x})}. \quad (1 \text{ mark})$$

Where $T(\underline{x}) = \sum_{i=1}^{n} \frac{(x_i-1)^2}{x_i}$. (1 mark).

Hence by the Neyman Factorisation Theorem $T(\underline{x}) = \sum_{i=1}^{n} \frac{(x_i-1)^2}{x_i}$ is a sufficient statistic since $g(T(\underline{x}),\theta)$ depends on θ only through $T(\underline{x})$.

Note: Q1 is unseen but similar to examples.

Q(b)) $f_X(x;\theta) = \dfrac{1}{\sqrt{2\pi x^3\theta}}\, e^{-\frac{(x-1)^2}{2x\theta}}$; $x>0$, $\theta>0$

$L(\theta) = \prod_{i=1}^{n} f_{X_i}(x_i;\theta)$.

$\log L(\theta) = \sum_{i=1}^{n} \log\left(f_{X_i}(x_i;\theta)\right)$

$= \sum_{i=1}^{n} \log\left(\dfrac{1}{\sqrt{2\pi x_i^3\theta}}\, e^{-\frac{(x_i-1)^2}{2x_i\theta}}\right)$.

$= \sum_{i=1}^{n}\left[-\frac{1}{2}\log(2\pi x_i^3) - \frac{1}{2}\log\theta - \dfrac{(x_i-1)^2}{2x_i\theta}\right]$

$= -\frac{n}{2}\log(2\pi) - \frac{3}{2}\sum_{i=1}^{n}\log x_i - \frac{n}{2}\log\theta - \frac{1}{2\theta}\sum_{i=1}^{n}\frac{(x_i-1)^2}{x_i}$ (1 mark).

$\dfrac{d\log L(\theta)}{d\theta} = -\dfrac{n}{2\theta} + \dfrac{1}{2\theta^2}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i}$

Equate $\dfrac{\log L(\theta)}{d\theta} = 0 \Rightarrow -\dfrac{n}{2\theta} + \dfrac{1}{2\theta^2}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i} = 0$

$\dfrac{n}{2\theta} = \dfrac{1}{2\theta^2}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i}$

$\theta = \dfrac{1}{n}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i}$ (1 mark).

$\dfrac{d^2\log L(\theta)}{d\theta^2} = \dfrac{d}{d\theta}\left[-\dfrac{n}{2\theta} + \dfrac{1}{2\theta^2}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i}\right]$

$= \dfrac{n}{2\theta^2} - \dfrac{1}{\theta^3}\sum_{i=1}^{n}\dfrac{(x_i-1)^2}{x_i}$ (1 mark)

At the point $\theta = \frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-1)^2}{x_i}$

$$\Rightarrow \sum_{i=1}^{n}\frac{(x_i-1)^2}{x_i} = n\theta$$

Hence $\frac{d^2 \log L(\theta)}{d\theta^2} = \frac{n}{2\theta^2} - \frac{1}{\theta^3}(n\theta)$. (1 mark).

$$= \frac{n}{2\theta^2} - \frac{n}{\theta^2} = -\frac{n}{2\theta^2} < 0.$$

indicating that the maximum likelihood estimator is $\hat{\theta}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-1)^2}{x_i}$

1c) Given that $E(X_i) = 1$ and $E\left(\frac{1}{X_i}\right) = 1+\theta$

$$E\left(\hat{\theta}_{MLE}\right) = E\left[\frac{1}{n}\sum_{i=1}^{n}\frac{(x_i-1)^2}{x_i}\right] = \frac{1}{n}E\left[\sum_{i=1}^{n}\left(\frac{x_i^2-2x_i+1}{x_i}\right)\right]$$ (1 mark).

$$= \frac{1}{n}E\left[\sum_{i=1}^{n}\left(x_i - 2 + \frac{1}{x_i}\right)\right]$$

$$= \frac{1}{n}E\left[\sum_{i=1}^{n}x_i - \sum_{i=1}^{n}2 + \sum_{i=1}^{n}\left(\frac{1}{x_i}\right)\right].$$

$$= \frac{1}{n}\left[\sum_{i=1}^{n}E(x_i) - 2n + \sum_{i=1}^{n}E\left(\frac{1}{x_i}\right)\right]$$ (1 mark).

$$= \frac{1}{n}\left[\sum_{i=1}^{n}(1) - 2n + \sum_{i=1}^{n}(1+\theta)\right]$$

$$= \frac{1}{n}\left[n - 2n + n + n\theta\right] = \theta$$ (1 mark)

$\therefore \hat{\theta}_{MLE}$ is an unbiased estimator of θ. (1 mark).

Q1(b)

1e) For large samples, we shall use the asymptotic properties of MLE.

$$\hat{\theta}_{MLE} \overset{\sim}{\approx} N\left(\theta, I^{-1}(\theta)\right)$$ (1 mark)

i.e. $$\hat{\theta}_{MLE} \overset{\sim}{\approx} N\left(\theta, \frac{2\theta^2}{n}\right)$$

$$\therefore \quad \frac{\hat{\theta}_{MLE} - \hat{\theta}}{\sqrt{\frac{2\theta^2}{n}}} = \frac{\sqrt{n}\left(\hat{\theta}_{MLE} - \theta\right)}{\theta\sqrt{2}} \xrightarrow{d} N(0,1)$$ (1 mark).

Therefore a $100(1-\alpha)\%$ C.I can be constructed as follows.

$$P\left(-Z_{\alpha/2} < \frac{\sqrt{n}\left(\hat{\theta}_{MLE} - \theta\right)}{\theta\sqrt{2}} < Z_{\alpha/2}\right) = 1-\alpha$$

$$P\left(-\frac{Z_{\alpha/2}\sqrt{2}}{\sqrt{n}} < \frac{\hat{\theta}_{MLE}}{\theta} - 1 < \frac{Z_{\alpha/2}\sqrt{2}}{\sqrt{n}}\right) = 1-\alpha$$

$$P\left(1 - Z_{\alpha/2}\sqrt{\frac{2}{n}} < \frac{\hat{\theta}_{MLE}}{\theta} < 1 + Z_{\alpha/2}\sqrt{\frac{2}{n}}\right) = 1-\alpha$$

Since $\theta > 0$

$$\Rightarrow P\left(\frac{\hat{\theta}_{MLE}}{1 + Z_{\alpha/2}\sqrt{\frac{2}{n}}} < \theta < \frac{\hat{\theta}_{MLE}}{1 - Z_{\alpha/2}\sqrt{\frac{2}{n}}}\right) = 1-\alpha$$

(2 marks).

If a double approximation is used by replacing θ with $\hat{\theta}_{MLE}$...

we would obtain

$$P\left(-z_{\alpha/2} < \frac{\sqrt{n}\,(\hat{\theta}_{MLE} - \theta)}{\hat{\theta}_{MLE}\sqrt{2}} < z_{\alpha/2} \right) = 1-\alpha.$$

$$P\left(-z_{\alpha/2}\sqrt{\frac{2}{n}} < 1 - \frac{\theta}{\hat{\theta}_{MLE}} < z_{\alpha/2}\sqrt{\frac{2}{n}} \right) = 1-\alpha.$$

$$P\left(-1 - z_{\alpha/2}\sqrt{\frac{2}{n}} < \frac{-\theta}{\hat{\theta}_{MLE}} < -1 + z_{\alpha/2}\sqrt{\frac{2}{n}} \right) = 1-\alpha.$$

since $\hat{\theta}_{MLE} > 0$

$$\Rightarrow P\left[-\hat{\theta}_{MLE}\left(1 + z_{\alpha/2}\sqrt{\frac{2}{n}}\right) < -\theta < -\hat{\theta}_{MLE}\left(1 - z_{\alpha/2}\sqrt{\frac{2}{n}}\right) \right] = 1-\alpha.$$

$$\Rightarrow P\left[\hat{\theta}_{MLE}\left(1 - z_{\alpha/2}\sqrt{\frac{2}{n}}\right) < \theta < \hat{\theta}_{MLE}\left(1 + z_{\alpha/2}\sqrt{\frac{2}{n}}\right) \right] = 1-\alpha.$$

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which is a symmetric interval.

i.e. $100(1-\alpha)\%$ C.I for θ is

$$\hat{\theta}_{MLE} \pm z_{\alpha/2}\,\hat{\theta}_{MLE}\sqrt{\frac{2}{n}}.$$

Question 2 —

a) $L(\theta; \underline{x}) = \prod_{i=1}^{n} \frac{1}{\Gamma(\alpha)\theta^{\alpha}} x_i^{\alpha-1} e^{-x_i/\theta}$, $\theta > 0$.

$= \left(\frac{\prod_{i=1}^{n} x_i^{\alpha-1}}{\Gamma(\alpha)^n} \right) \theta^{-\alpha n} e^{-\frac{1}{\theta}\sum_{i=1}^{n} x_i}$ $\theta > 0$.

2 marks (1 if minor errors), unseen but similar to examples

b) Derive MLE (similar to examples)

$$\log L(\theta; \underline{x}) = -\alpha n \log(\theta) - \frac{1}{\theta}\sum_{i=1}^{n} x_i$$

$$\frac{d}{d\theta} \log L(\theta; \underline{x}) = -\frac{\alpha n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i = 0 \quad \text{1 mark}$$

$\Rightarrow \hat{\theta}(\underline{x}) = \frac{\sum_{i=1}^{n} x_i}{\alpha n}$ 1 mark

From CRLB corollary

Alternatively $\frac{d}{d\theta} \log L(\theta; \underline{x}) = -\frac{\alpha n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i = \underbrace{\frac{\alpha n}{\theta^2}}_{I(\theta)} \left(\underbrace{\frac{\sum_{i=1}^{n} x_i}{\alpha n}}_{\hat{\theta}(\underline{x})} - \theta \right)$

c) Students can show $I(\theta) = \frac{n\alpha}{\theta^2}$ by applying the definition, or directly by using the CRLB corollary, as above. (2 marks, similar to examples)

$E\{\hat{\theta}(\underline{x})\} = \theta$ ✓ unbiased

$Var\{\hat{\theta}(\underline{x})\} = \frac{\theta^2}{\alpha n}$ ✓ efficient.

d) Students can derive $E(\hat{\theta}(\underline{x}))$ and $Var(\hat{\theta}(\underline{x}))$ by using properties of gamma r.v.s, by solving integrals, or by using the CRLB corollary, as above.

(unseen, but similar to examples, 2 marks for bias, 2 marks for variance, 1 mark if minor errors).

(more challenging material)

e) $Y_1, \dots, Y_n$ related to $X_1, \dots, X_n$ by

$$Y_i = \gamma_i X_i \quad , \quad \gamma_i > 0 \qquad \forall i \in \{1, \dots, n\}.$$

Because $X_i \sim \text{Gamma}(\alpha, \theta)$

$$Y_i \sim \text{Gamma}(\alpha, \gamma_i \theta)$$

by using the scaling property of the gamma r.v.

$$f(\underline{y}; \theta) = \prod_{i=1}^{n} f(y_i; \theta)$$

$$= \prod_{i=1}^{n} \frac{1}{\Gamma(\alpha) \gamma_i^{\alpha} \theta^{\alpha}} y_i^{\alpha-1} e^{-y_i/\gamma_i\theta}$$

note that obs. are indep. but not ident. distributed.

$$= \frac{1}{\Gamma(\alpha)^n} \left(\prod_{i=1}^{n} \frac{y_i^{\alpha-1}}{\gamma_i^{\alpha}} \right) \theta^{-\alpha n} e^{-\frac{1}{\theta} \sum_{i=1}^{n} y_i/\gamma_i}$$

(2 marks for correctly calculating the likelihood)

By following the steps of part (b) we can show that the MLE

$$\hat{\theta}(\underline{Y}) = \frac{\sum_{i=1}^{n} Y_i/\gamma_i}{\alpha n}$$ (1 mark for MLE)

is the MVUE for this problem. Moreover, it attains the CRLB, $\text{Var}(\hat{\theta}(Y)) = \frac{\alpha n}{\theta^2}$. (2 marks for comment)

Question 3 →

unseen, but similar to class examples

a) $$f(\underline{x};p) = \prod_{i=1}^{m} \binom{n}{x_i} p^{x_i}(1-p)^{x_i - n}$$

$$= \left(\prod_{i=1}^{m} \binom{n}{x_i}\right) p^{\sum x_i} (1-p)^{nm - \sum_{i=1}^{m} x_i}$$

From the NP Lemma, the most powerful test of size α (1 mark)

has reject region $R = \{x : \frac{f(\underline{x};p_0)}{f(\underline{x};p_1)} < k_\alpha\}$ 1 mark

with k_α adjusted so that

$P_r(\underline{X} \in R \mid p = p_0) = \alpha$.

$$R = \left\{x : \left(\frac{p_0}{p_1}\right)^{\sum_{i=1}^{m} x_i} \left(\frac{1-p_0}{1-p_1}\right)^{nm - \sum_{i=1}^{m} x_i} < k_\alpha\right\}$$ 1 mark

unseen, but similar to examples

b) We note that the LR is a monotonically decreasing function of the statistic $T(\underline{x}) = \sum_{i=1}^{m} x_i$ 1 mark

$\left(\frac{p_0}{p_1}\right)^{\sum_{i=1}^{m} x_i}$ (< 1) $\left(\frac{1-p_0}{1-p_1}\right)^{nm - \sum_{i=1}^{m} x_i}$ (> 1) 1 mark

So this problem admits a UMP test with

$$R = \left\{x : \sum_{i=1}^{m} x_i > k'_\alpha\right\}$$ 1 mark

with k'_α such that the test has size α. This test is equivalent to the test of part (a).

c) $m = 4$, $n = 5$, $p_0 = 0.4$, $\alpha = 0.06$ bookwork

$$\left(\sum_{i=1}^{m} x_i\right) \sim \text{Bin}\left(\frac{nm}{20}, p_0\right)$$ ($p_0 = 0.4$) 1 mark

We find on the table that

$k'_\alpha = 13$ 2 marks for correct k_α

bookwork

d) The power of the test is given by $P\left[\sum_{i=1}^{m} X_i > 13 \,\middle|\, p = 0.8\right]$ — 1 mark

p_1

By using the table we find that the power is $\approx 91\%$.

This is a moderate power, to improve it we would more data (i.e. larger n). — 1 mark — 1 mark

Similar to examples

e) The p-value for $t = T(\underline{x}) = \sum_{i=1}^{m} x_i = 17$ is the probability of observing $T(\underline{X}) \geq t$. — 1 mark

This probability is 0.01. — 1 mark

This is a small p-value, it indicates a very low degree of confidence on the null hypothesis. — 1 mark

Question 4

* Question 4 is bookwork and follows closely class exercises.

a) PDF, $f(\theta) \propto \exp\left(\frac{-(\theta-\mu)^2}{2\sigma_2^2}\right)$

$$\propto \exp\left(\frac{-(\theta^2-2\mu\theta)}{2\sigma_2^2}\right) \quad -(1)$$

Likelihood,

$$p(\underset{\sim}{x}|\theta) \propto \prod_{i=1}^{n} \exp\left(\frac{-(x_i-\theta)^2}{2\sigma_1^2}\right)$$

$$= \exp\left[\frac{1}{2\sigma_1^2} \times \sum_{i=1}^{n}(x_i-\theta)^2\right]$$

$$\propto \exp\left[\frac{-1}{2\sigma_1^2}(n\theta^2-2n\theta\bar{x})\right] \quad (1)$$

$$p(\theta|\underset{\sim}{x}) \propto \left[\exp\left(\frac{-(\theta^2-2\mu\theta)}{2\sigma_2^2}\right)\right] \times \left[\exp\left(\frac{-(n\theta^2-2n\theta\bar{x})}{2\sigma_1^2}\right)\right]$$

$$= \exp\left[\frac{n\theta\bar{x}}{\sigma_1^2} - \frac{n\theta^2}{2\sigma_1^2} + \frac{\mu\theta}{\sigma_2^2} - \frac{\theta^2}{2\sigma_2^2}\right] \quad (1)$$

This is proportional to

$$\exp\left(\frac{-(\theta-\mu_*)^2}{2\sigma_*^2}\right) \propto \exp\left(\frac{-\theta^2}{2\sigma_*^2} + \frac{\mu_*\theta}{\sigma_*^2}\right)$$

Comparing these two will give us

$$\sigma_*^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + n\sigma_2^2} \quad -①$$

And, $$\mu_* = \frac{\mu\sigma_1^2 + n\sigma_2^2 \bar{x}}{\sigma_1^2 + n\sigma_2^2} \quad -①$$

Posterior, $\theta|\underset{\sim}{x} \sim N(\mu_*, \sigma_*^2)$

- b) Bayesian estimate of θ under quadratic loss
  = mean of the posterior

$$E(\theta|\underset{\sim}{x}) = \mu_* = \frac{\mu\sigma_1^2}{\sigma_1^2 + n\sigma_2^2} + \frac{n\sigma_2^2 \bar{x}}{\sigma_1^2 + n\sigma_2^2} \quad ①$$

c) $$E(\theta|\underset{\sim}{x}) = \frac{\sigma_1^2}{\sigma_1^2 + n\sigma_2^2}\mu + \frac{n\sigma_2^2}{\sigma_1^2 + n\sigma_2^2}\bar{x}$$

- $$= (1-Z)\mu + Z\bar{x} \quad ①$$

which is in the form of a credibility estimate with

$$Z = \frac{n\sigma_2^2}{\sigma_1^2 + n\sigma_2^2} \quad ①$$

which is the credibility factor.

d)

	Company A	Company B
n	5	5
σ_1^2	500	350
σ_2^2	800	600
$Z = \frac{n\sigma_2^2}{\sigma_1^2 + n\sigma_2^2}$	0.8889 (1)	0.8955 (1)
$\bar{x}$	439	356 (1)
μ	400	300
Credibility Premium $= (1-Z)\mu + Z\bar{x}$	434.67 (1)	350.15 (1)

e) $\sigma_1^2 \uparrow \Rightarrow Z \downarrow$ and credibility premium moves towards prior mean.

$\sigma_2^2 \uparrow \Rightarrow Z \uparrow$ and credibility premium moves towards sample mean.

1 mark each